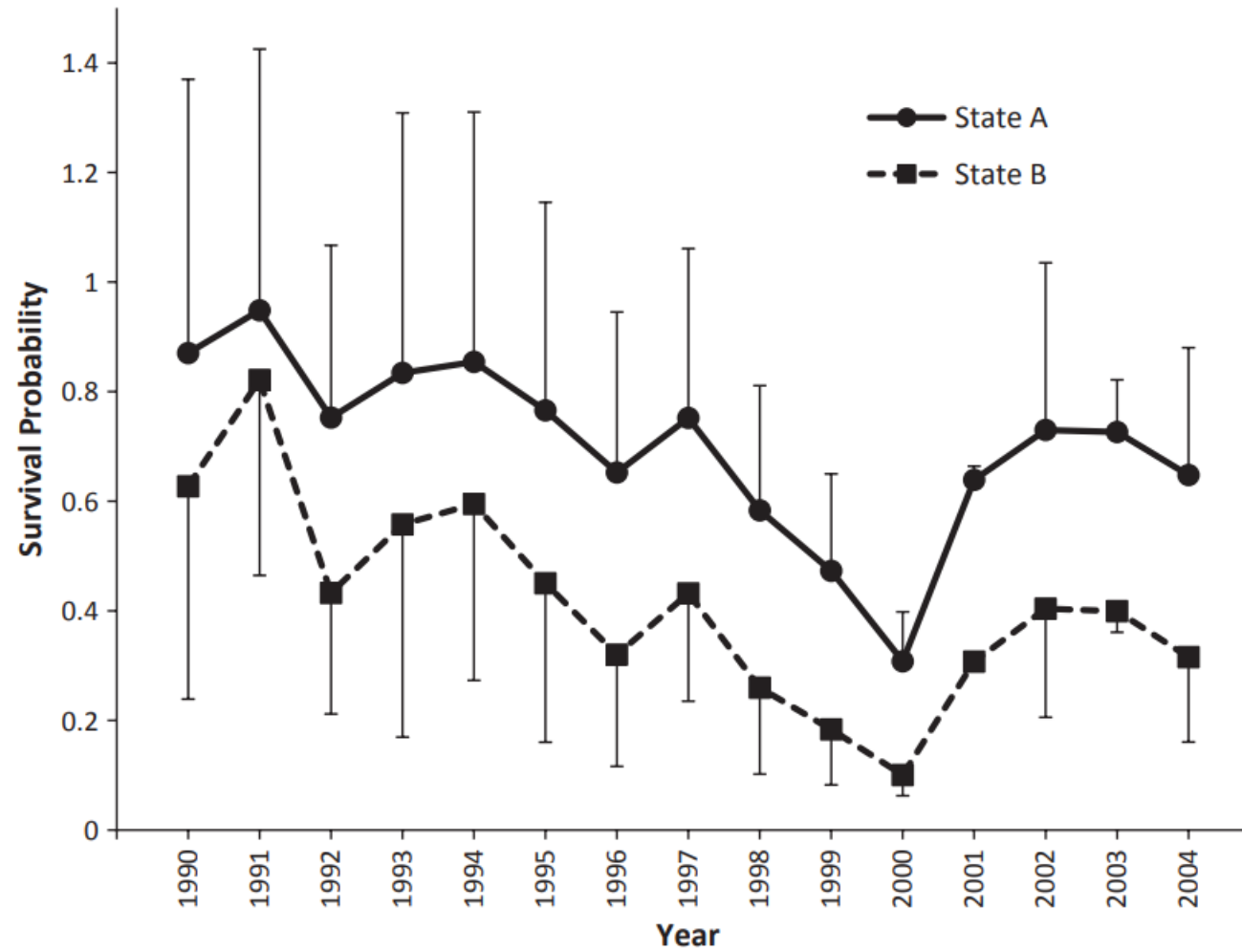


# Link functions: constraining parameter space to the possible



## But first, let's talk about 'assignments'

|        |                  |                                                                       |                              |
|--------|------------------|-----------------------------------------------------------------------|------------------------------|
| Week 3 | Sep 10<br>Sep 12 | linear models: lm() vs. brms() vs. custom<br>Random and fixed effects | Potential proposal topic due |
|--------|------------------|-----------------------------------------------------------------------|------------------------------|

Table 2. Grading.

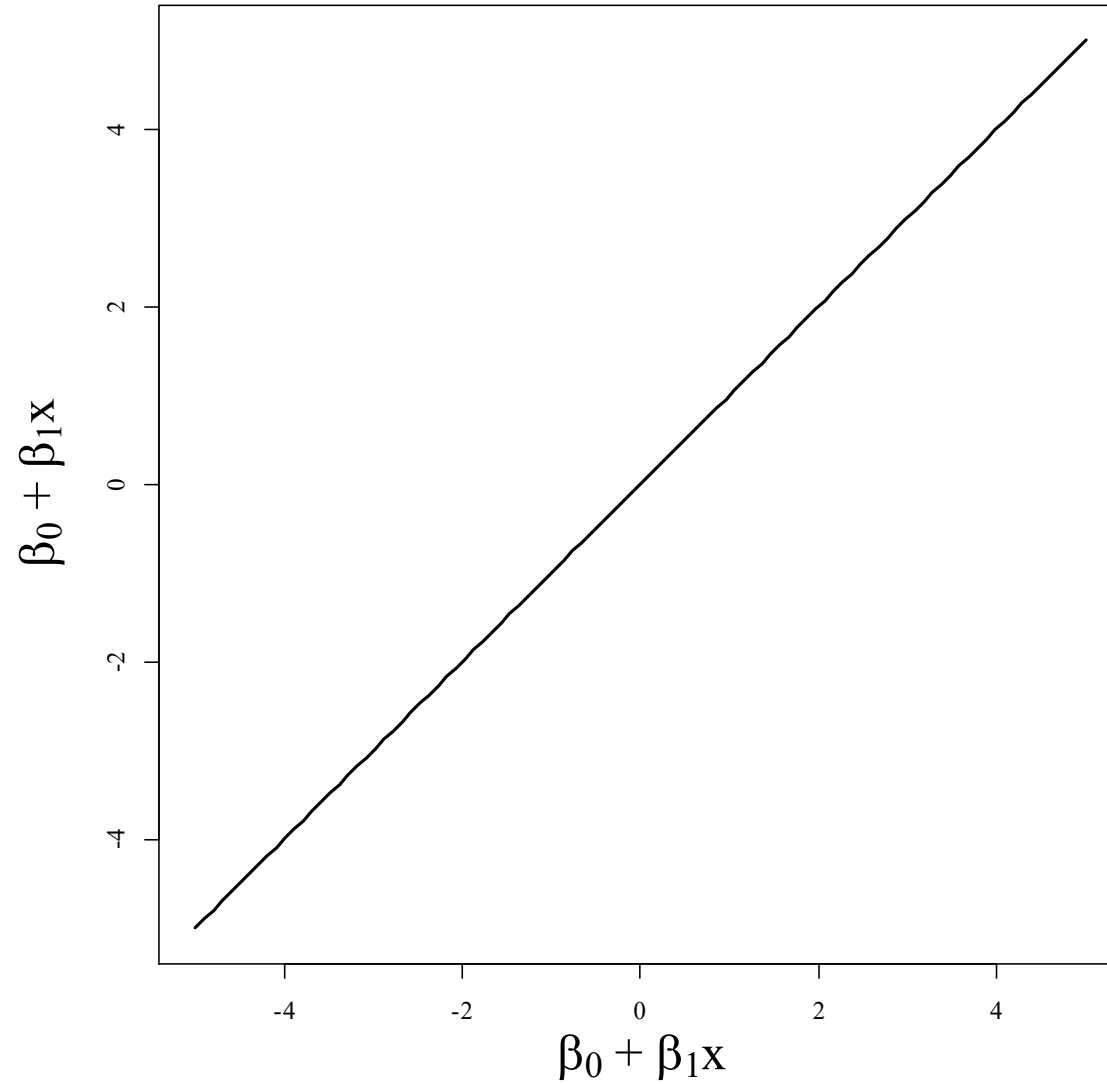
| Category      | Description                         | Date | Points | %  |
|---------------|-------------------------------------|------|--------|----|
| Project       | Potential topic                     |      | 10     | 5  |
|               | Final topic                         |      | 10     | 5  |
|               | Final project                       |      | 80     | 40 |
| Homeworks     |                                     |      | 60     | 30 |
| Participation | Contributing to discussion and labs |      | 40     | 20 |
| Total         |                                     |      | 200    |    |

And a promise...

**Thus far we've discussed linear models**

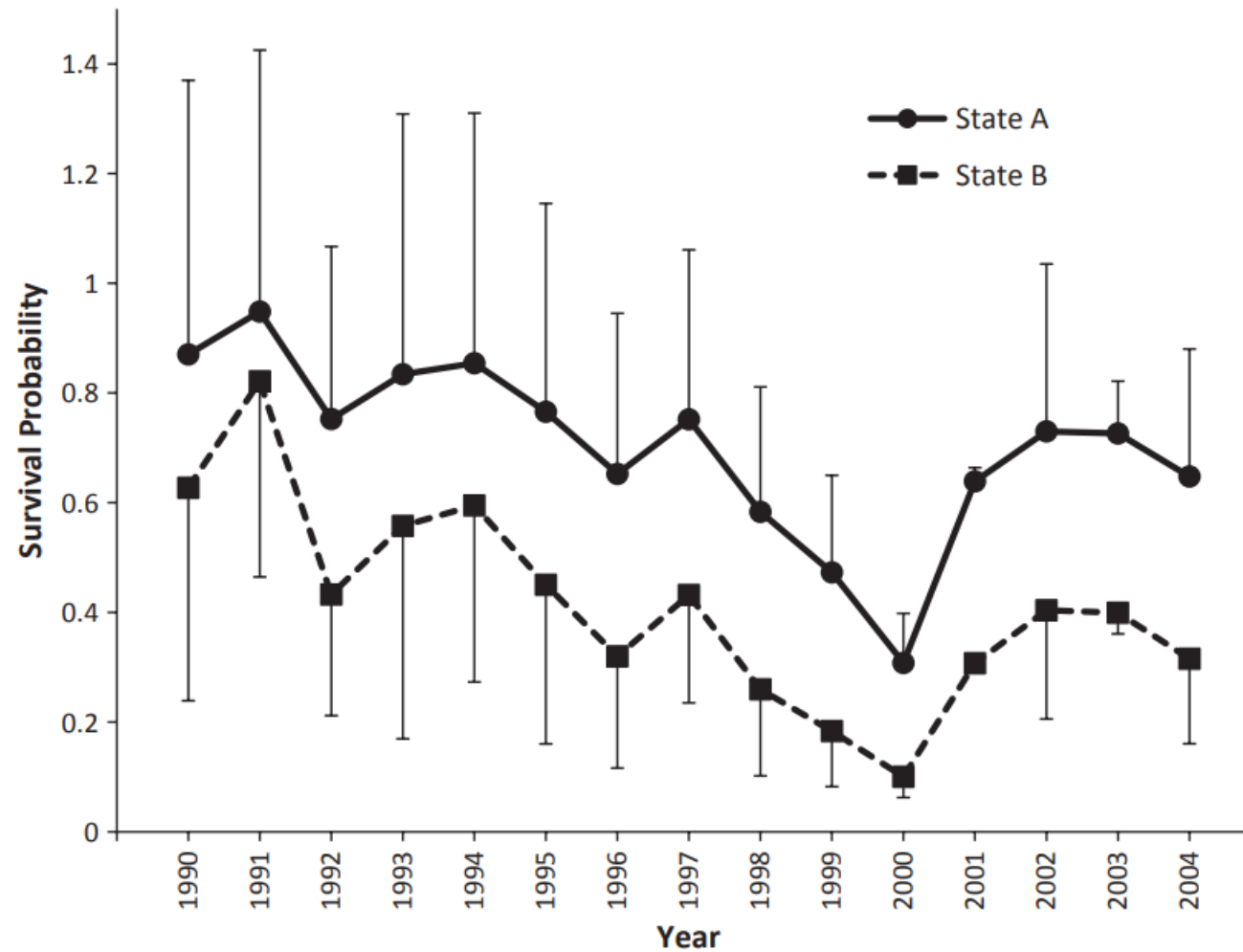
# Linear models use an 'identity link'

The scale of our response variable  $[-\infty, \infty]$



The scale of our model  $[-\infty, \infty]$

# Transformations are often necessary [take a close look]



Lindberg [43, 4755], Sedinger [53; 7668], and Lebreton [76; 22357] **(2013)** *Ecology and Evolution*

**Transformations are often necessary**

**How do we prevent ourselves from saying crazy [stuff]?**

**Let's imagine some ecological data types**

# Let's imagine some ecological data types

1. Counts of individuals [discrete, never less than 0?!]
2. Lengths, weights, or other morphometric measurements [continuous, never less than 0?]
3. Number of eggs or offspring [discrete, never less than 0]



# Let's imagine some ecological data types

1. Proportion or percent cover of something or other
2. Survival [yes/no; 1/0]
3. Pregnancy [yes/no; 1/0]
4. Hatching [yes/no; 1/0]
5. Morph [categorical?; 1/0 or 1, 2, ... , n]

# **All of these data types share some common attributes**

1. You can't have  $< 0$ 's
2. Many can only be 0's or 1's

**We often need to constrain parameters to be  $> 0$  or between 0 and 1.**

**So how do we handle that in our models?**

**Link functions!**

**How the heck do we do that?**

**Link functions: constraining parameter space to the possible**



**Or, what is a natural log?**

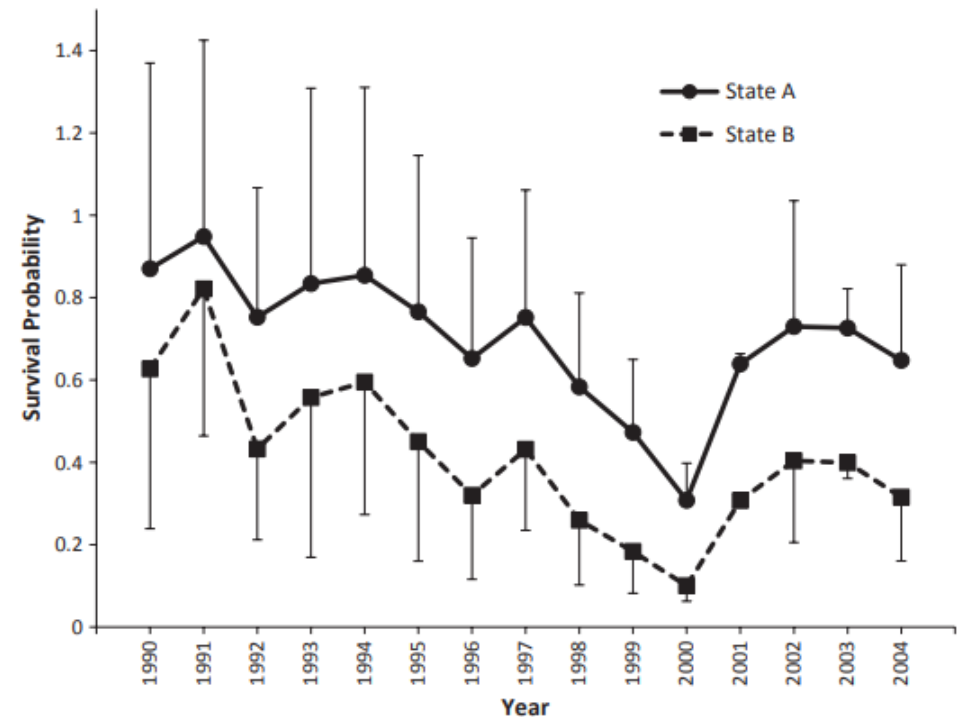
**What is  $e$ ? [an irrational and transcendental number]**

# e is the basis of many of the link functions used in Ecology

Link functions allow us to constrain parameter estimates so we don't predict things like 150% survival, 200% shrub cover, or that an alligator that is -2 ft long will weigh -100 pounds

M. S. Lindberg *et al.*

Heterogeneity and Harvest Dynamics



**Figure 2.** First-year survival probability of brant for states A and B. Errors bars are  $\pm 1$  standard error in single direction for clarity.

# What is e? [an irrational and transcendental number]

Imagine you have a bank account with \$1 in it that earns 100% interest [put more money in the account!]

If you calculate interest annually, at the end of the year, you'll have \$2

$$\$1 \times 2^1 = \$2$$



# What is e? [an irrational and transcendental number]

Imagine you have a bank account with \$1 in it that earns 100% interest [put more money in the account!]

If you calculate interest biannually, at the end of the year, you'll have \$2.25

$$\$1 \times 1.5^2 = \$2.25$$

# What is e? [an irrational and transcendental number]

Imagine you have a bank account with \$1 in it that earns 100% interest [put more money in the account!]

If you calculate interest quarterly, at the end of the year, you'll have \$2.44

$$\$1 \times 1.25^4 = \$2.44$$

# What is $e$ ? [an irrational and transcendental number]

Compounding annually yields 2

Compounding biannually yields 2.25

Compounding quarterly yields 2.4414...

Compounding monthly yields 2.613035....

Compounding weekly yields 2.692597...

Compounding daily yields 2.714567...

**Notice a pattern?**

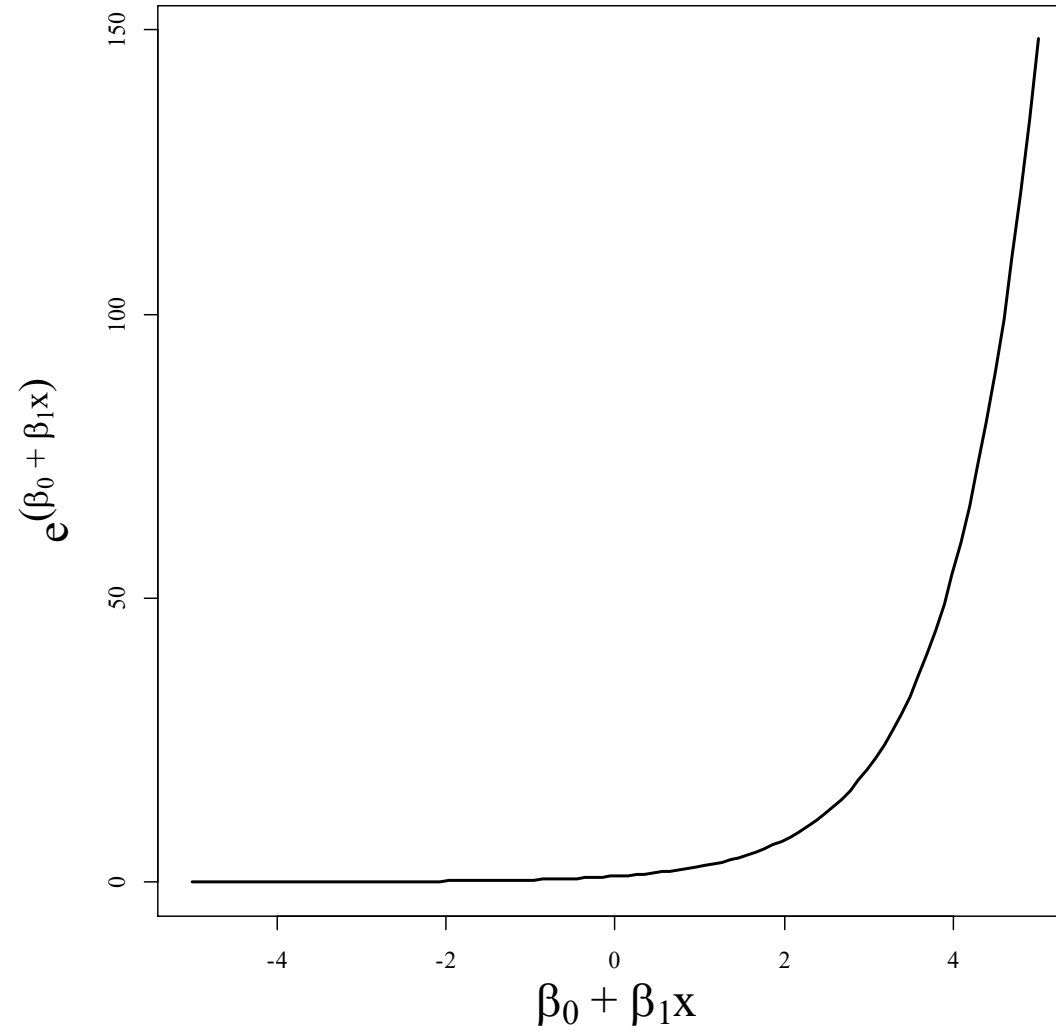
$$e = \lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right)^n$$

Compounding **continuously** yields e

2.71828182845904523536028747135266249775724709369995957496696762772  
 407663035354759457138217852516642742746639193200305992181741359662  
 904357290033429526059563073813232862794349076323382988075319525101  
 901157383418793070215408914993488416750924476146066808226480016847  
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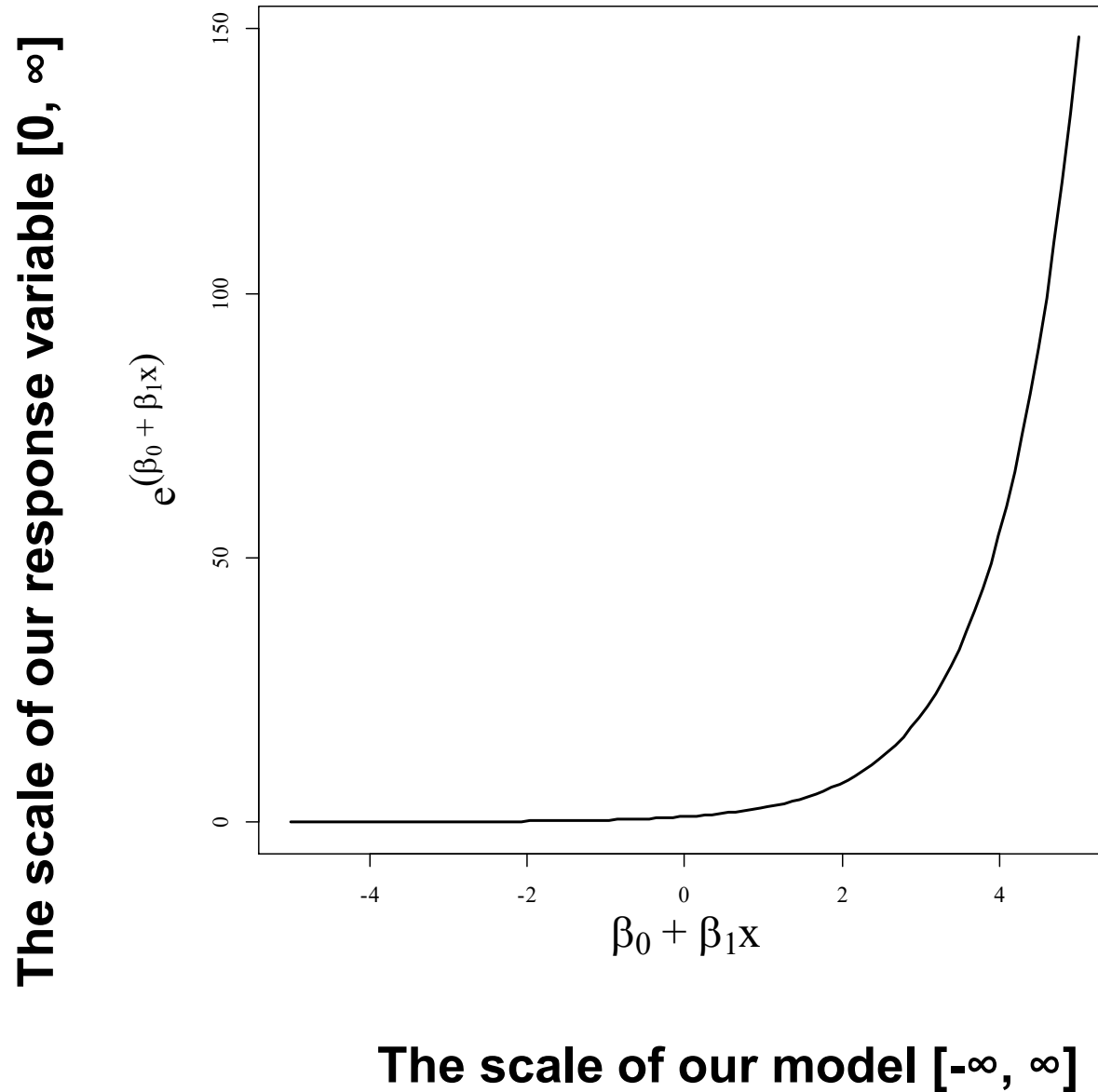
**We can do beautiful things with e (constrain to  $> 0$ )**

**The scale of our response variable  $[0, \infty]$**



**The scale of our model  $[-\infty, \infty]$**

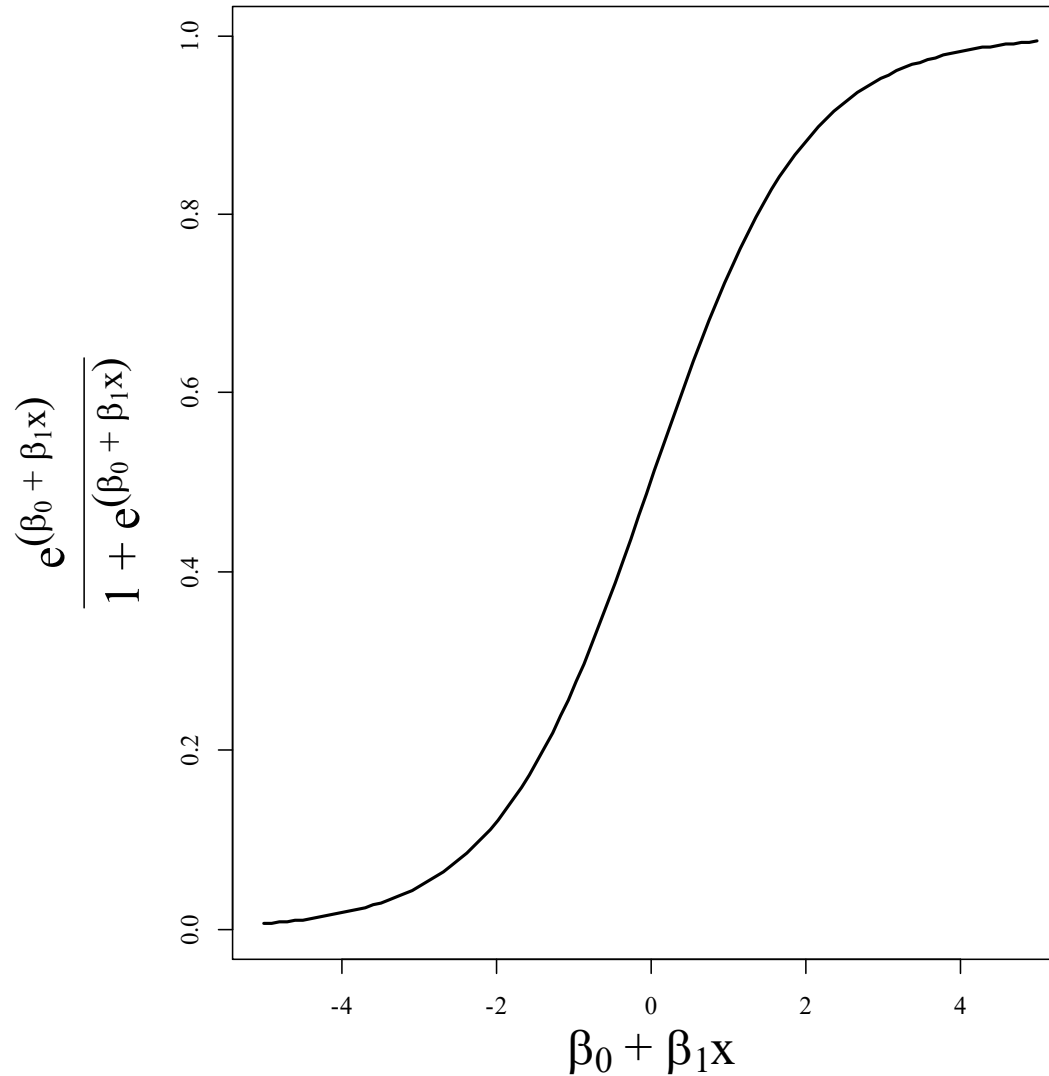
**We can do beautiful things with e (constrain to  $> 0$  &  $< 1$ )**



$$y = e^{\beta_0 + \beta_1 \times x}$$
$$\beta_0 + \beta_1 \times x = \ln(y)$$

**We can do beautiful things with e (constrain to  $> 0$  &  $< 1$ )**

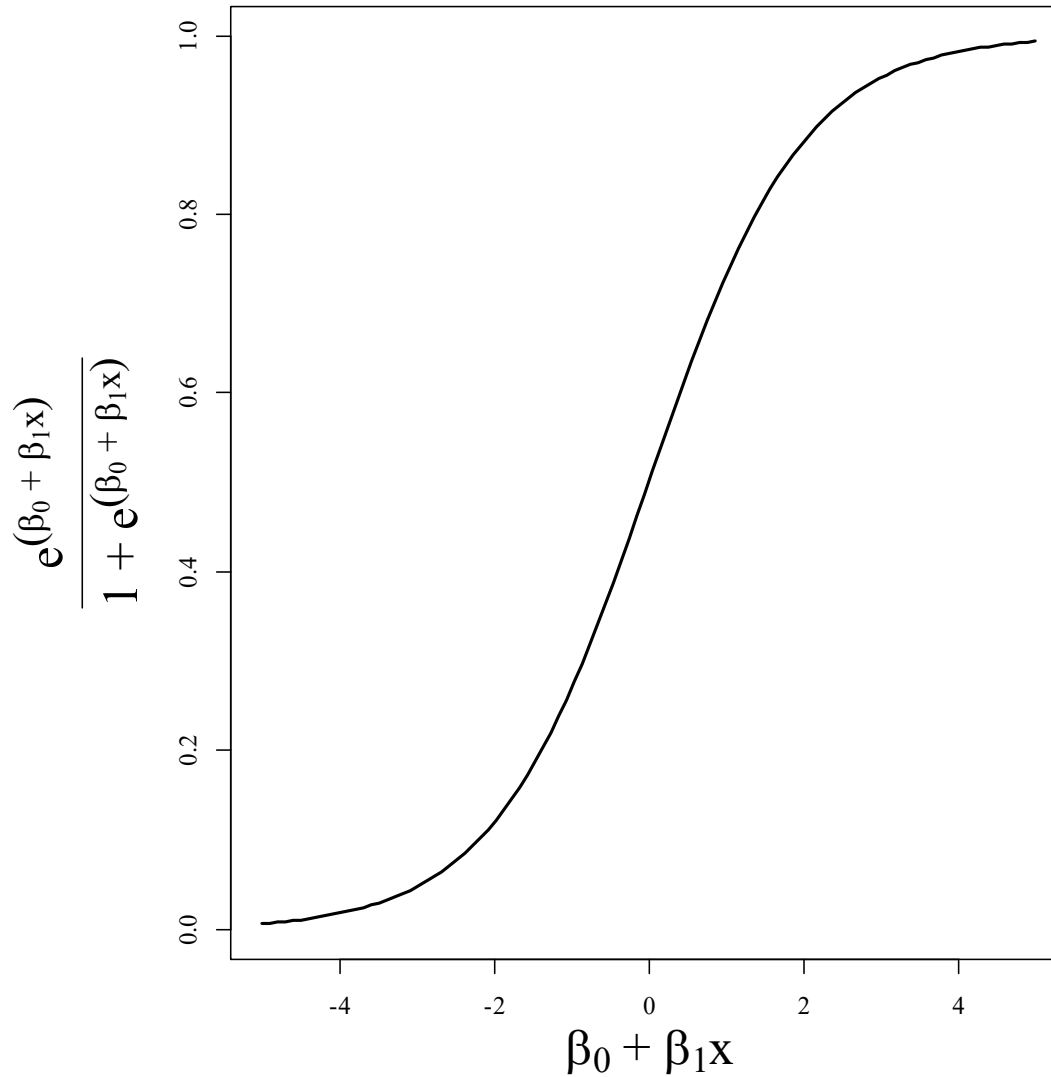
**The scale of our response variable  $[0, 1]$**



**The scale of our model  $[-\infty, \infty]$**

**We can do beautiful things with e (constrain to  $> 0$  &  $< 1$ )**

**The scale of our response variable [0, 1]**



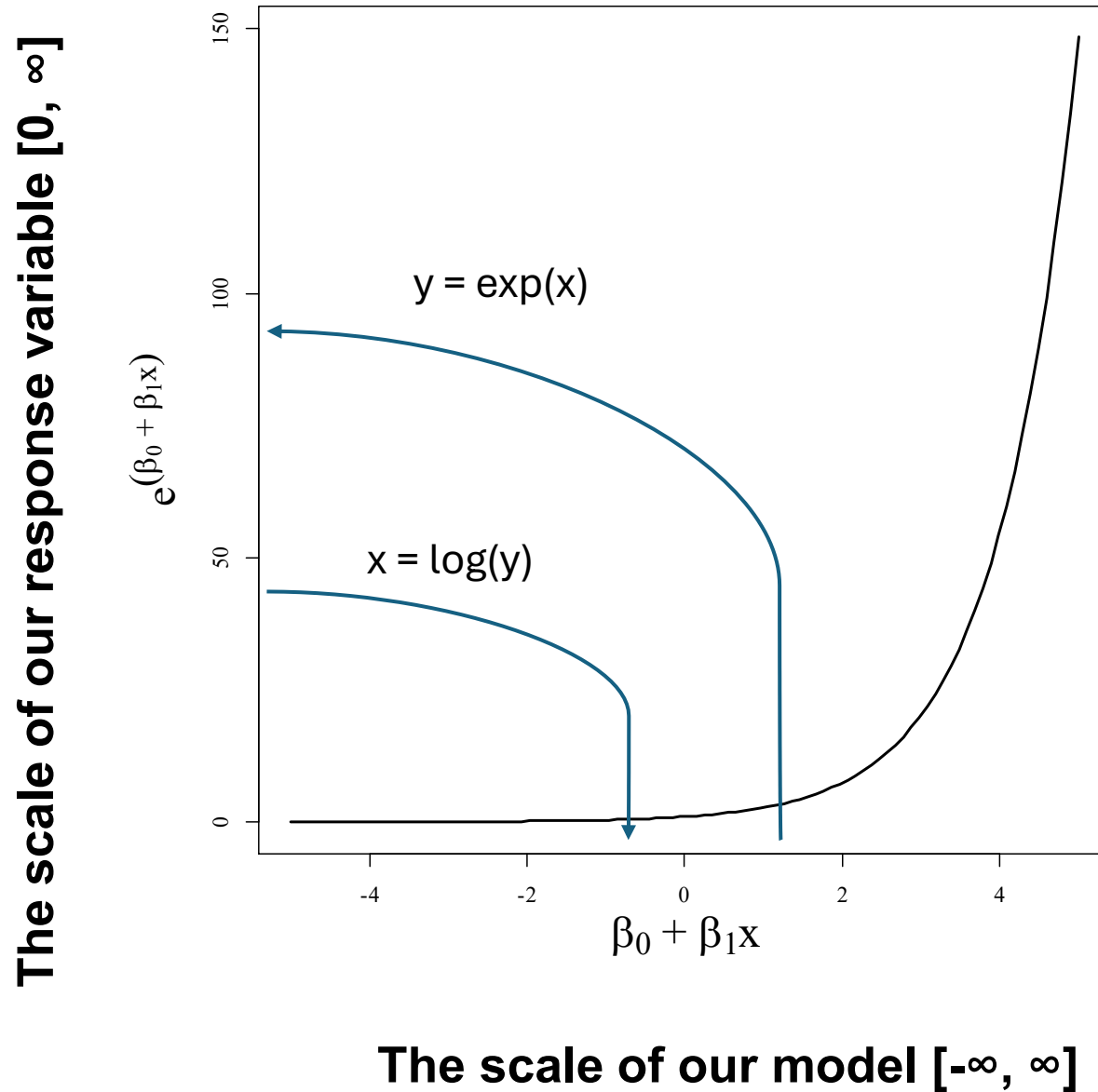
**The scale of our model  $[-\infty, \infty]$**

$$y = \frac{e^{\beta_0 + \beta_1 \times x}}{1 + e^{\beta_0 + \beta_1 \times x}}$$

$$\beta_0 + \beta_1 \times x = \ln\left(\frac{y}{1 - y}\right)$$



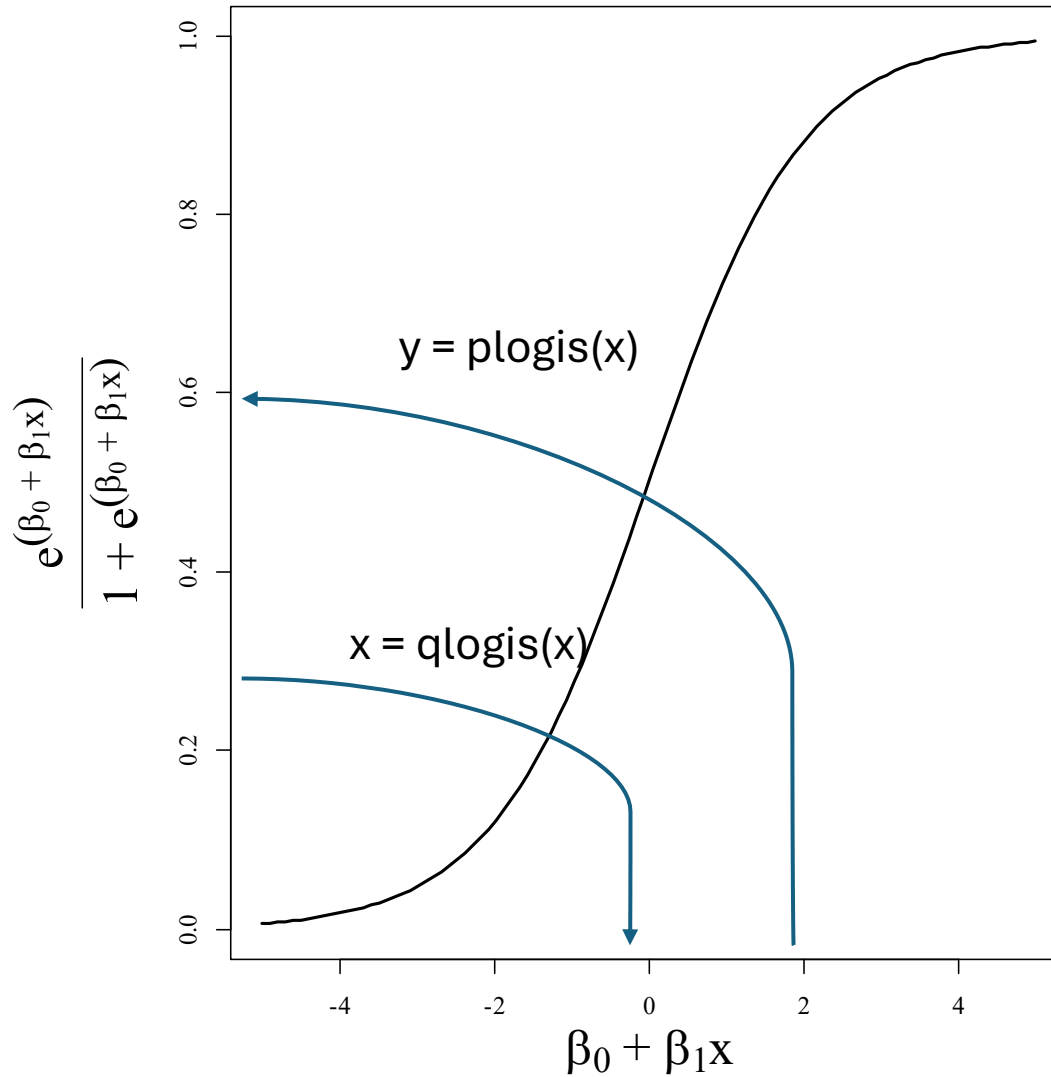
# Implementing in base R (log link)



$$y = e^{\beta_0 + \beta_1 \times x}$$
$$\beta_0 + \beta_1 \times x = \ln(y)$$

# Implementing in base R (logit link)

The scale of our response variable [0, 1]



The scale of our model  $[-\infty, \infty]$

$$y = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

$$\beta_0 + \beta_1 x = \ln\left(\frac{y}{1 - y}\right)$$

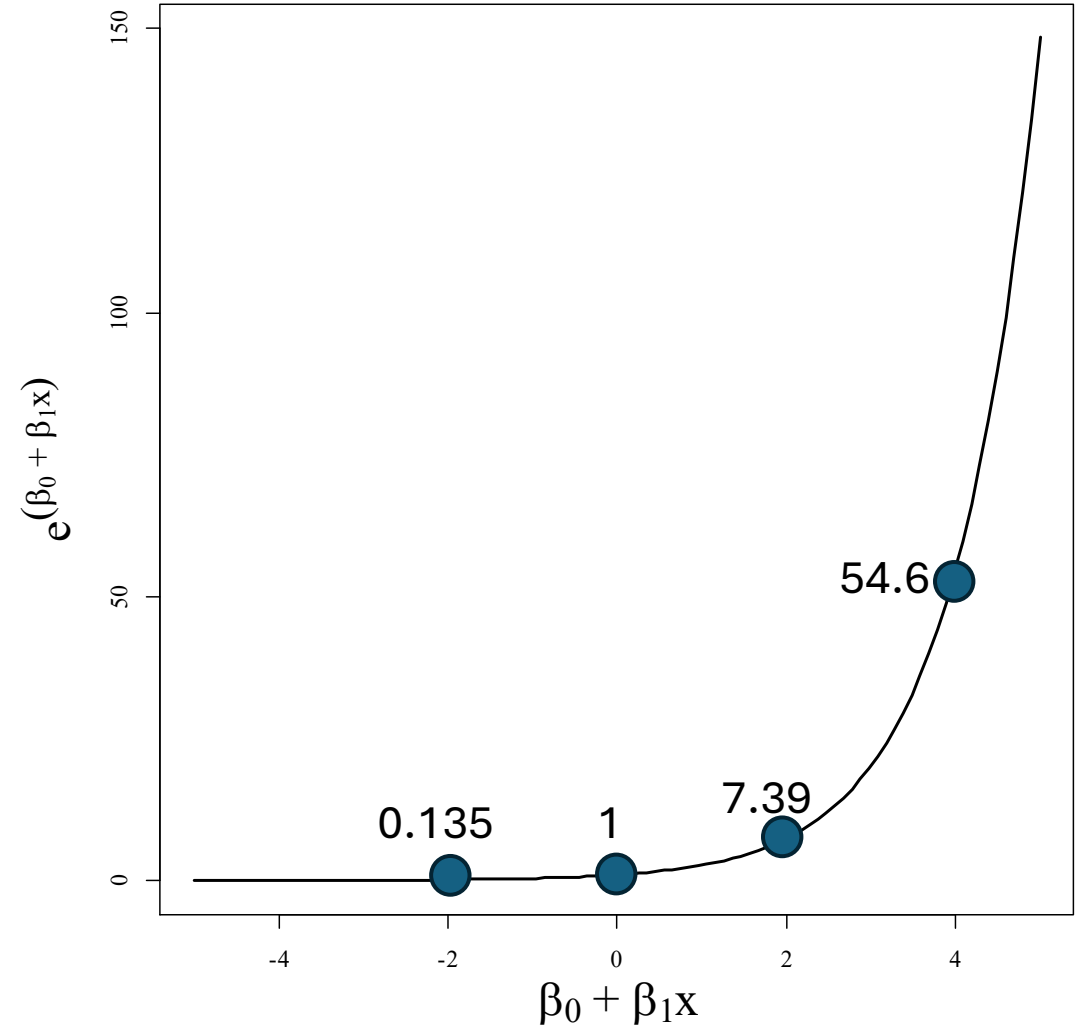
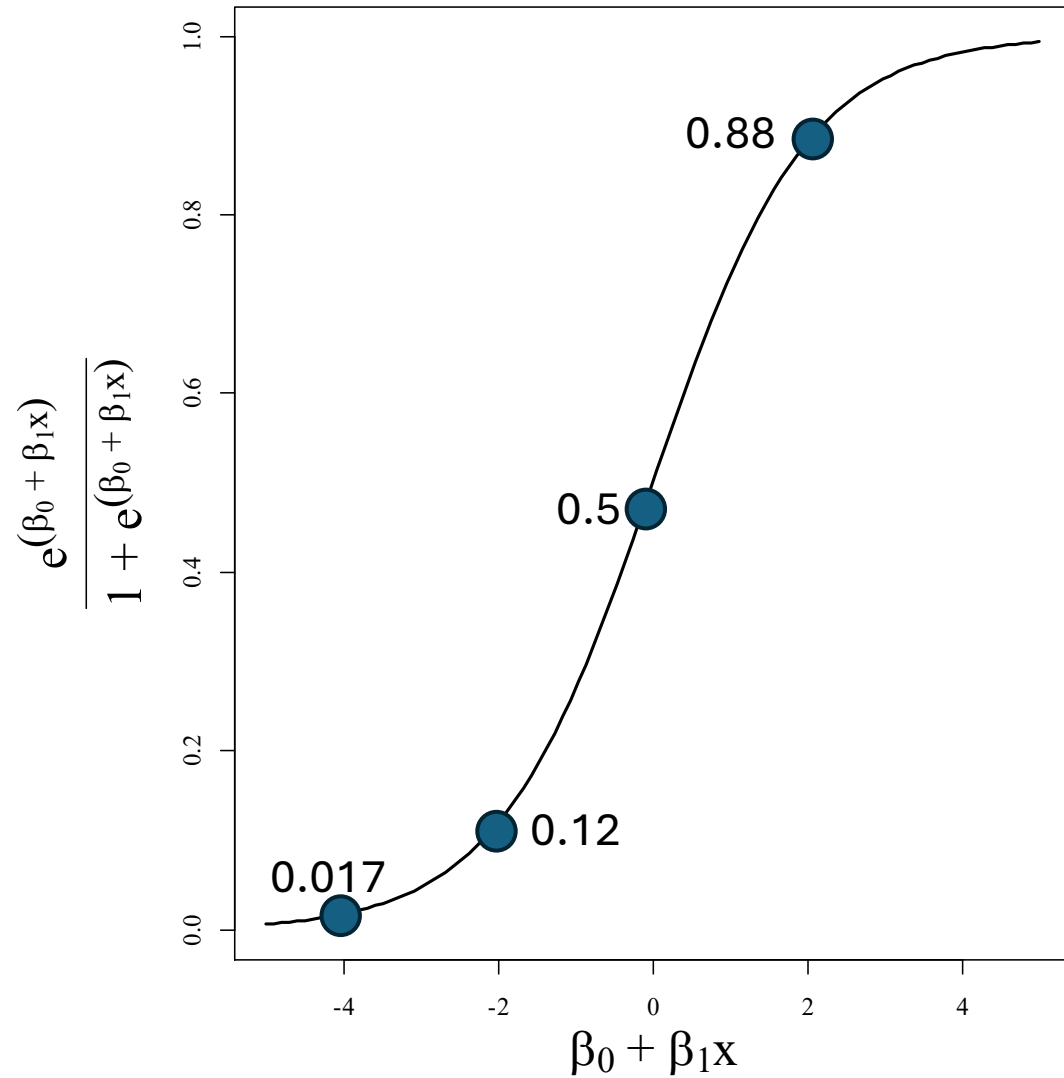
# Today, we'll reanalyze two datasets...

1. We'll model the alligator mass  $\sim$  length data using a log-normal distribution (i.e., constraining mass to be positive!)
2. We'll model some duck band-recovery data using a logit-link (i.e., constraining the probability of being shot, recovered, and reported to be  $> 0$  and  $< 1$ )

## Things to think about:

1. Beta estimates are no longer 'cut and dry' rise/run. They are relative to your position on the link function and the shape of the link function.

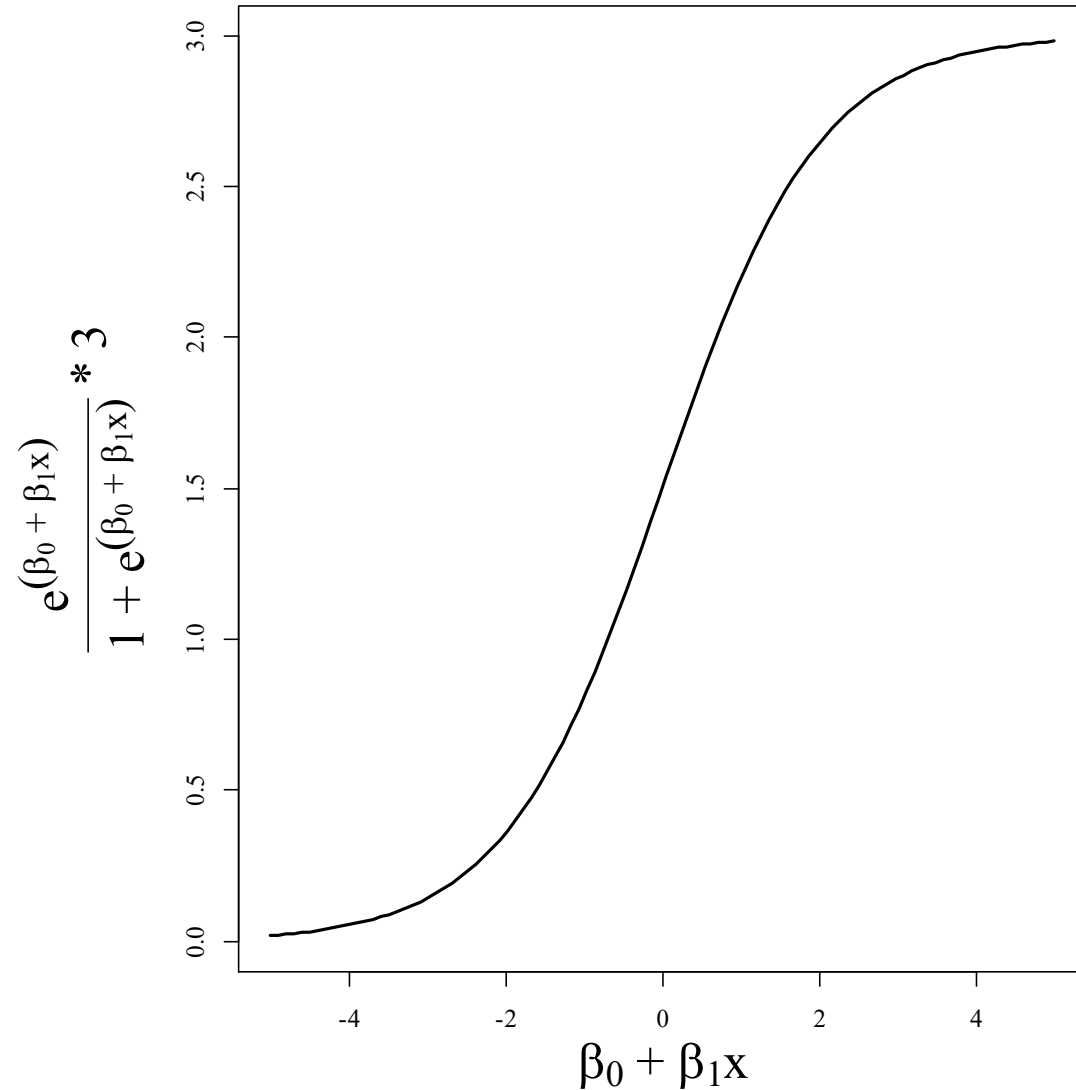
# Beta estimates are no longer cut and dry!



## Things to think about:

1. Beta estimates are no longer 'cut and dry' rise/run. They are relative to your position on the link function and the shape of the link function.
2. There are many, many different types of link functions that do very similar things. Sometimes using a slightly different function can help with parameter estimation (related to point #1).

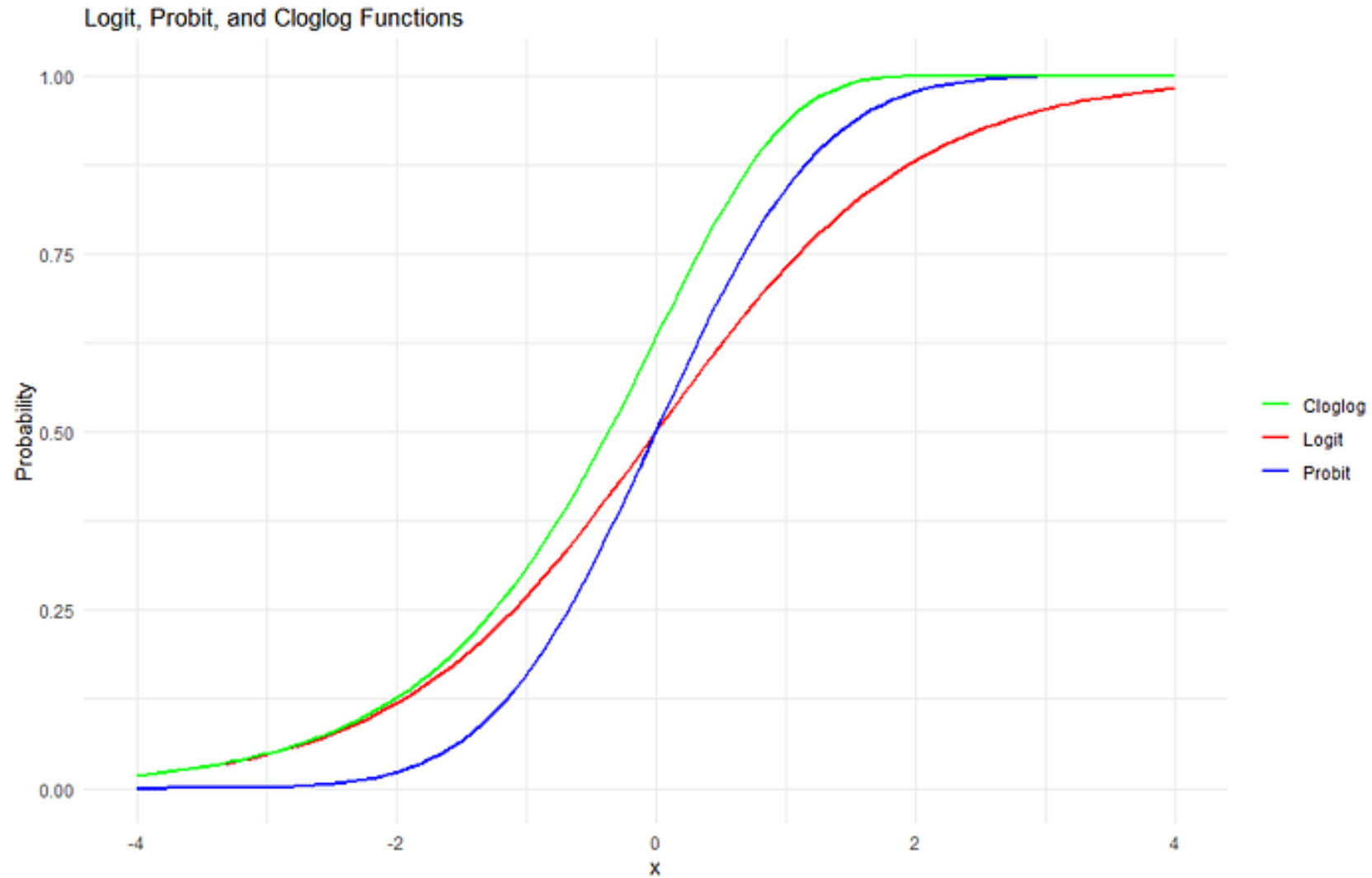
## Different link functions (an example to constrain b/w 0 and 3)



$$y = 3 \times \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

$$\beta_0 + \beta_1 x = \ln \left( \frac{\frac{y}{3}}{1 - \frac{y}{3}} \right)$$

# Different link functions (different shapes)

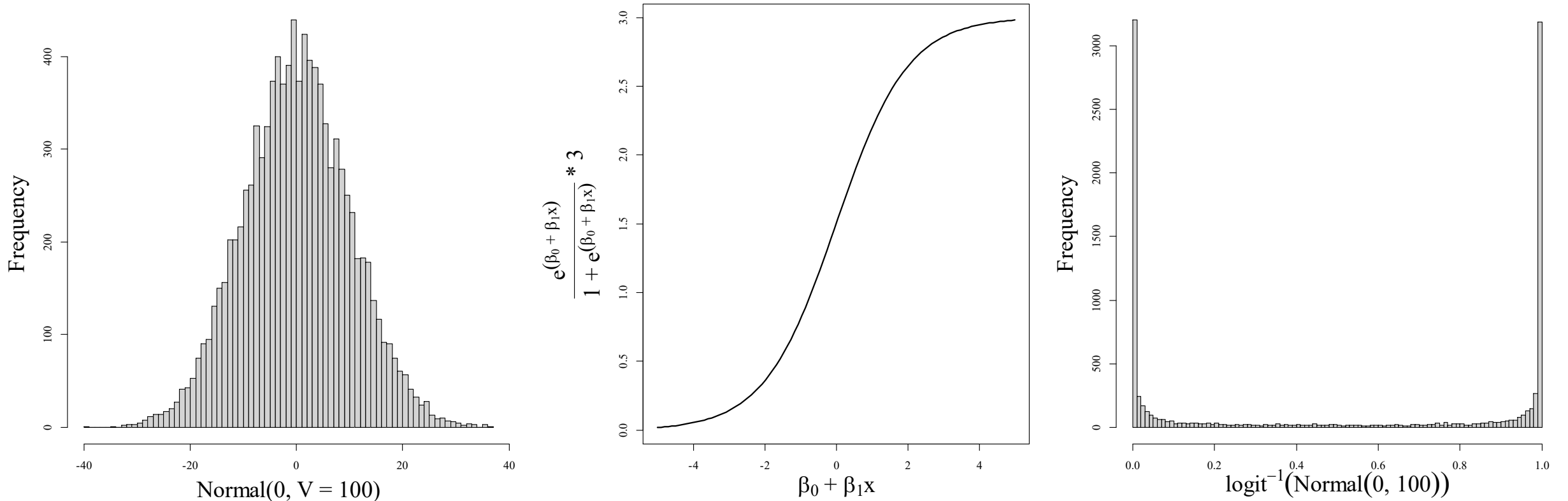




## Things to think about:

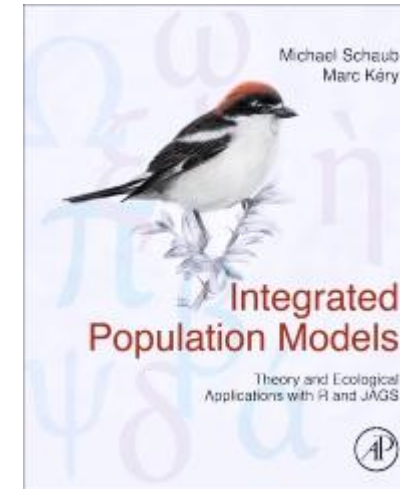
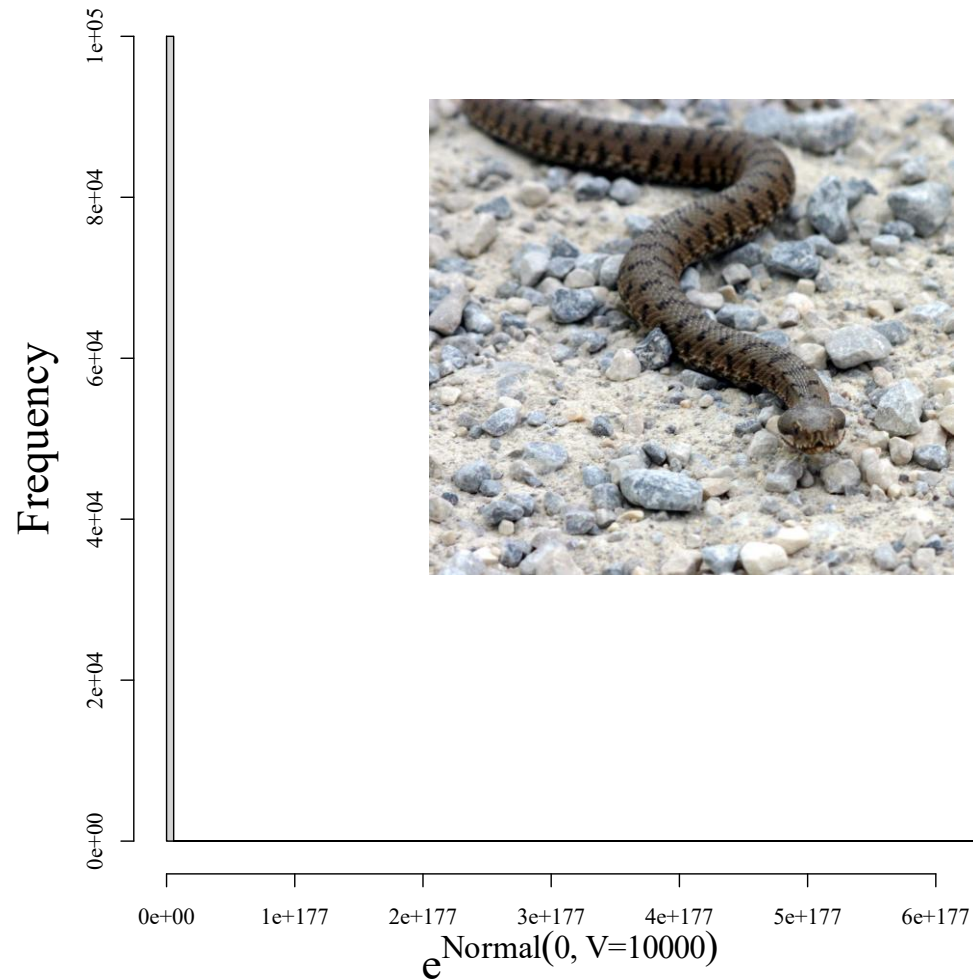
1. Beta estimates are no longer 'cut and dry' rise/run. They are relative to your position on the link function and the shape of the link function.
2. There are many, many different types of link functions that do very similar things. Sometimes using a slightly different function can help with parameter estimation (related to point #1).
3. Priors suddenly become **quite a bit more complex** to think about... (see Northrup and Gerber 2018 for a logit-link example... and Lecture 2 for some log-link examples, e.g., snakes larger than a solar system!)

# Priors on link functions



Northrup & Gerber (2018) *PLOS ONE*

# Priors on link functions

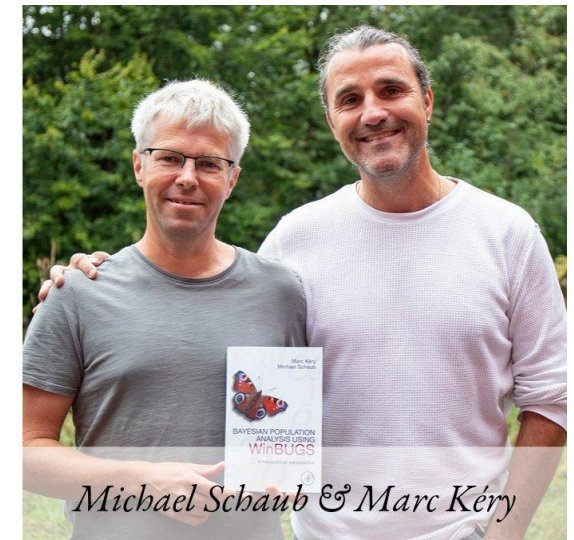


## *The Marsh Award for* **INNOVATIVE ORNITHOLOGY**

*Nominated*

Michael and Marc were nominated for their groundbreaking work on Bayesian hierarchical models, changing the way we use statistics to analyse large, citizen-science data sets. The methods have not only helped BTO, but are being used worldwide on a variety of data sets and applications. Their books, as well as their workshops and teaching, only adds value to their work.

MARSH  
Christian Trust



*Michael Schaub & Marc Kéry*

**Earth's diameter is 12,756 km**